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Designing an AI-Integrated Automation System for Post Offices

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1.1 Introduction – Business Context



Characteristics 1

Vietnam Post operates in an environment that requires the continuous receipt and dispatch of orders across the country.



Characteristics 2

The acceptance stage is the initial point where data for the entire postal journey is generated. The quality of data created at this stage directly impacts subsequent processes.



Characteristics 3

Over-the-counter transactions still require multiple manual processing steps and data entry tasks performed directly by tellers, resulting in significant time consumption.



Characteristics 4

Volume spikes by 25–30% during peak periods (e.g., Lunar New Year 2026), leading to severe counter congestion.

Business Context (Vietnam Post)

VIETNAM POST

Vietnam Post is a leading state-owned enterprise in the fields of postal services, delivery, postal financial services, and logistics.

1.2 Introduction – Problem Statement

Prolonged Acceptance Time: Manual process takes 6–8 minutes per item

Repetitive Data Entry: Information already exists on package labels, but tellers must still manually read and re-type it into the system.

High Risk of Errors: Manual reading and entry of physical measurements (weight, dimensions) easily cause data errors, impacting downstream logistics.

Teller resources are currently consumed by mechanical tasks that can be automated: Tellers' time should shift from reading, typing, recording measurements, and performing repetitive calculations to inspecting goods, providing advice, serving customers, cross-selling services, and handling exceptions.

Goal

Develop a solution capable of supporting the automation of the parcel intake process, specifically regarding information recognition, package dimension measurement, and the calculation of necessary parameters.

2.1 Background

Computer Vision Application Trends

In recent years, Computer Vision and Deep Learning have become core technologies in logistics automation and parcel handling. Modern systems go beyond basic sorting, aiming for end-to-end automation—ranging from information recognition and 3D dimension measurement to parcel condition inspection. Research directly related to this topic focuses on three main problem categories:

- Text and handwriting recognition (OCR) on parcel labels.
- Parcel dimension measurement and volume estimation.
- Multi-task integration on a single computer vision system.

Literature Review

2.2 Related Works

Li et al. – Text Recognition with TrOCR

Architectural Innovation: Replaces traditional CNN + RNN combinations with a Transformer-based Image Encoder and Text Decoder.

Peng et al. – Dimension Measurement via Laser Triangulation & Deep Learning

Combines laser triangulation with deep learning on a structured light system to extract 3D depth features.

Fernandes et al. – Dimension Measurement via Geometry and Laser

Determines box dimensions from camera images based on perspective geometry, object contours, and supporting laser lines.

Le Van Thai – Dimension Measurement via Stereo Vision

Uses a dual-camera system (Stereo Vision) observing from two different angles, calculating the disparity of corresponding points to reconstruct 3D space.

Tran et al. – Integrated Processing System with YOLOv12

Develops an integrated system based on YOLOv12-BiFPN-ResCBAM for three tasks: barcode reading, dimension measurement, and damaged parcel detection.

2.2 Related Works

Author	Method	Task	Advantages	Disadvantages
Li et al.	TrOCR (Transformer)	Printed/handwritten text OCR	High accuracy, end-to-end	Requires language-specific fine-tuning
Fernandes et al.	Perspective geometry + Laser	2D/3D dimension measurement	Clear mathematical foundation, automated	Dependent on laser hardware
Peng et al.	Laser Triangulation + DL	Depth & volume measurement	High 3D accuracy	Expensive equipment costs
Le Van Thai	Stereo Vision (2 Cameras)	Classification & 3D measurement	Non-contact measurement	Sensitive to lighting and occlusion
Tran et al.	YOLOv12-BiFPN-ResCBAM	Barcode, Dimension, Defect detection	Multi-task, optimized for edge devices	Leaves handwritten OCR unaddressed

2.3 Research Gap & Proposed Solution

Research Gap

- **Cost vs. Accuracy**: High-precision measurement methods using geometry + lazer or lazer + deep learning are constrained by the high cost of expensive laser and depth camera equipment.
- **Lack of Comprehensive Multi-Tasking**: No existing system simultaneously integrates physical dimension measurement and Vietnamese handwritten text OCR to extract package information.
- **Data Structuring Problem**: Raw OCR lacks a label region classification workflow (Layout Analysis) to convert extracted text into structured data (Full Name, Address, Phone Number).

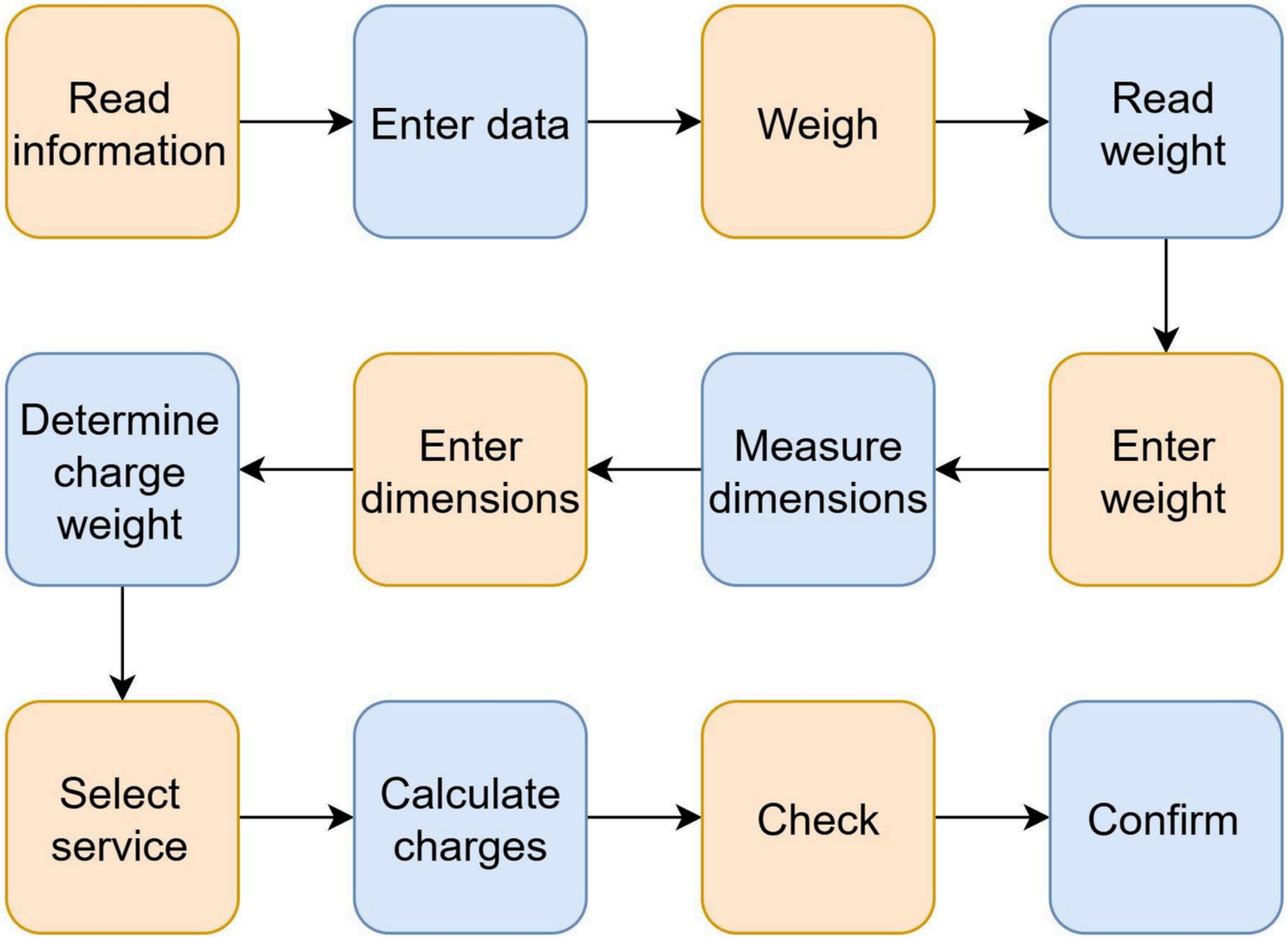
Proposed Solution

- **Integrated Multi-Tasking**: Develop an end-to-end pipeline: Parcel Detection ➔ Dimension/Volume Measurement ➔ Vietnamese Handwritten Text OCR.
- **Hardware Optimization**: Utilize YOLO + Canny Edge + 2 standard cameras for dimension measurement, completely eliminating costly laser and depth camera hardware.
- **Fine-Tuning Vietnamese TrOCR**: Specialize the TrOCR model on realistic handwritten text data from Vietnamese postal shipping forms.

3.1 Business Process Analysis

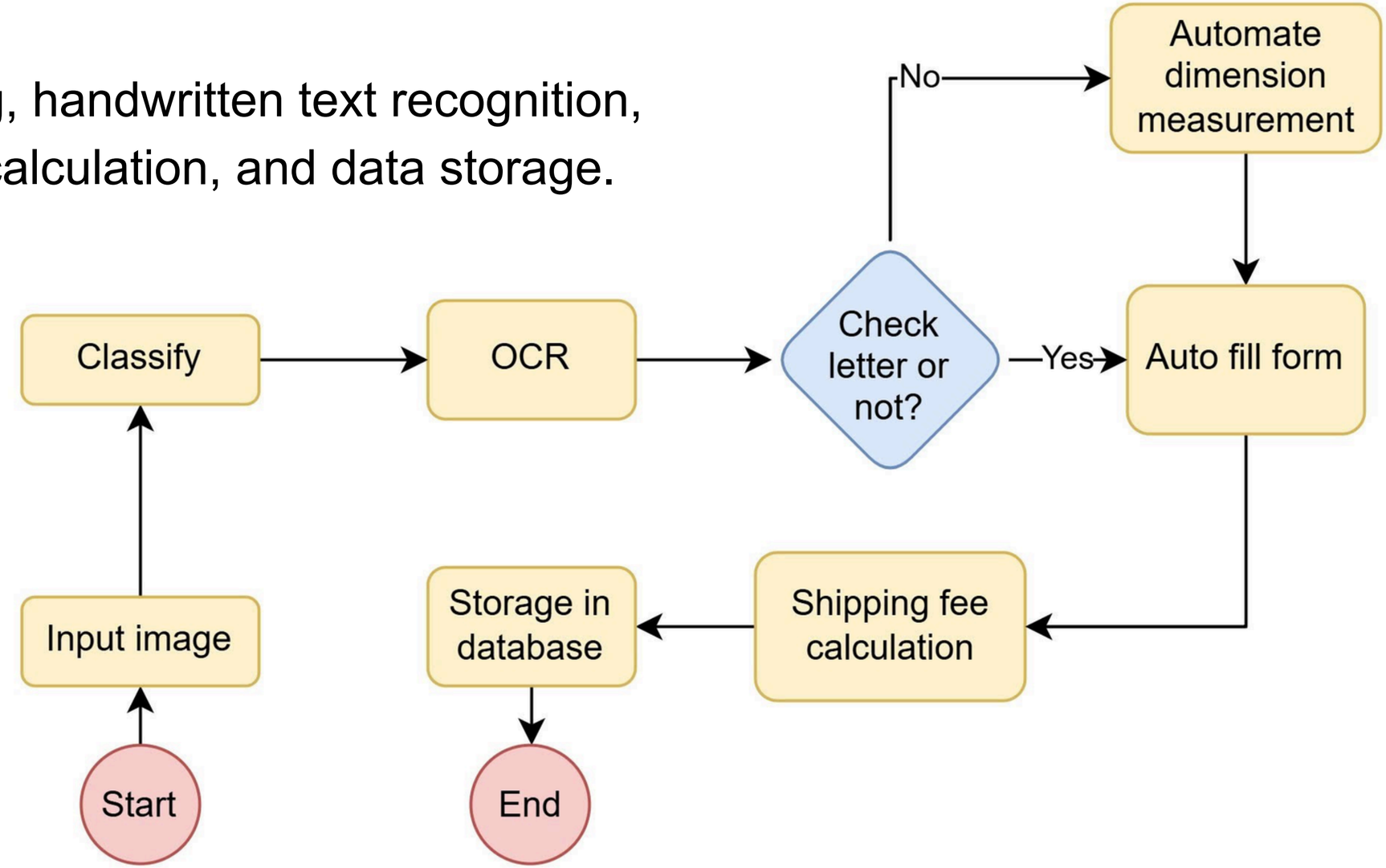
Current process

The current system involves multiple small steps that must be performed manually, resulting in significant time consumption (6–8 minutes per order) and a high risk of errors.



3.2 Overall Framework

System supporting sorting, handwritten text recognition, volume calculation, fee calculation, and data storage.



3.3 Handwriting Extraction and Recognition Module

1. Classify
Envelope vs. Parcel

YOLOv11n

2. Localize
Information-label region

YOLOv11n

3. Detect
Text-region bounding boxes

EasyOCR (detection only)

4. Recognize
Handwritten characters

Fine-tuned TrOCR

5. Normalize
Raw text into structured JSON

Claude API

03 Methodology

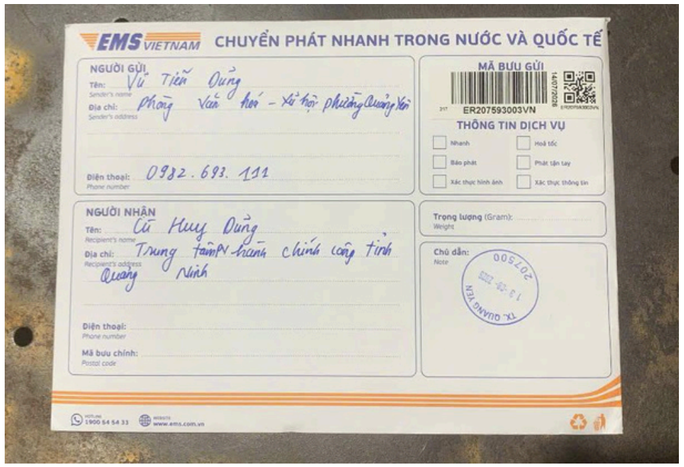
3.3.1 Classification & Information-Region Localization

Model 1: Envelope/parcel classification

Purpose: Processes original input images and classifies postal items into two main categories: Box or Envelope.

Dataset preparation:

- Target Classes: Filtered to keep 2 core categories (Label-Box, Label-Envelope).
- Data Augmentation: Expanded training set to 1,050 samples using vertical flip, ±15° rotation, and ±15° brightness adjustments.
- Image Resizing: All images resized to 640 × 640.
- Dataset Partitioning:
  - Training Set: 1,050 samples
  - Validation Set: 44 samples
  - Testing Set: 44 samples





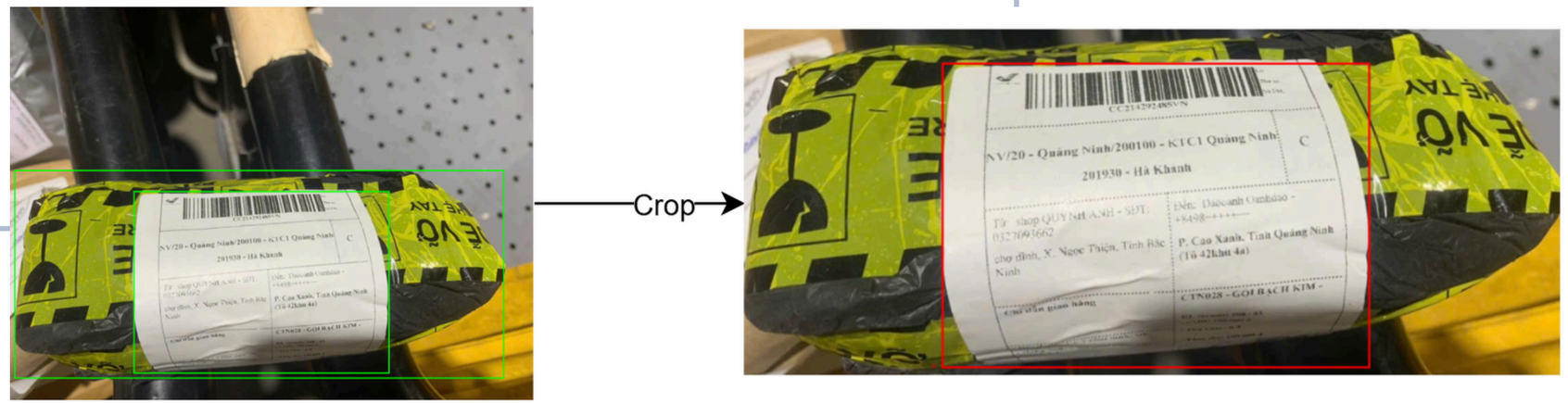
3.3.1 Classification & Information-Region Localization

Model 2: Information-region localization

Purpose: Detects the exact shipping label / information region (Info Region) on cropped box surfaces for downstream OCR processing.

Dataset preparation:

- Source Selection: Selected box-only images from the dataset; envelope images were excluded as Model 2 focuses exclusively on box labels.
- Auto-Cropping: Cropped box regions from raw images using bounding box coordinates to isolate target surfaces.
- Single Class Objective: Fine-tuned to detect a single target class (Info Region).
- Dataset Partitioning:
  - Training Set: 1,050 samples
  - Validation Set: 44 samples
  - Testing Set: 44 samples



3.3.1 Classification & Information-Region Localization

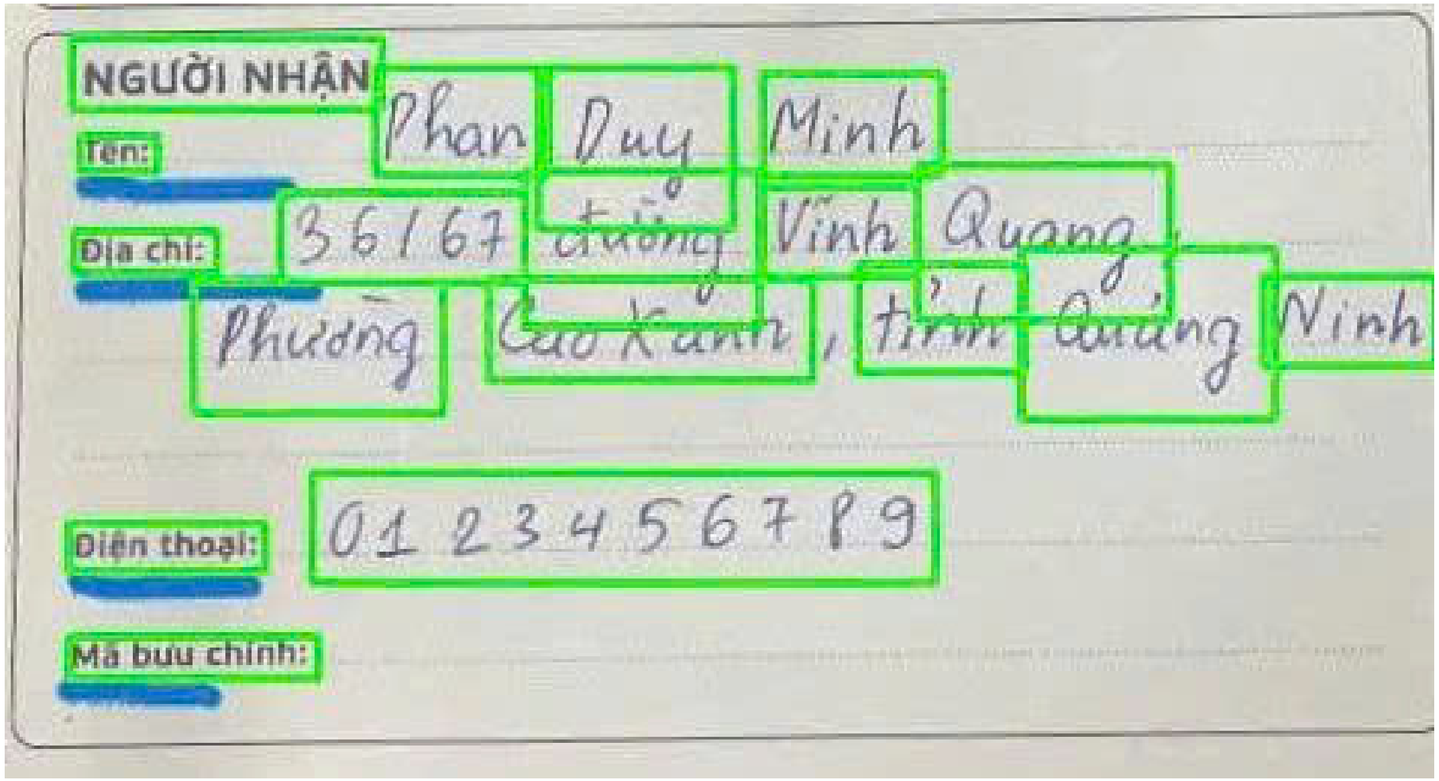
Model & Training Environment

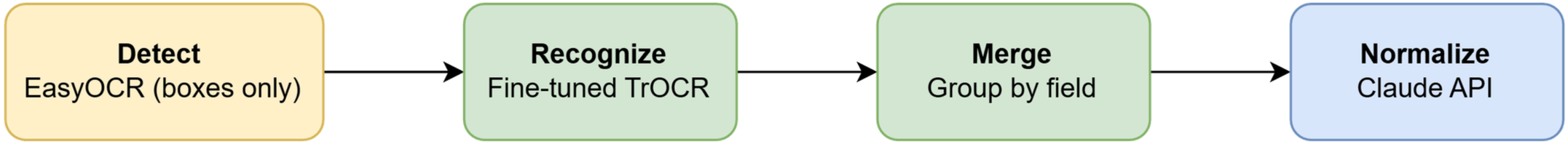
Shared Configuration: This training preset and hyperparameter setup is applied identically to both Model.

- Base Model: YOLO11n (Nano architecture, optimized for lightweight real-time execution).
- Pretrained Weights: Fine-tuned from COCO pretrained weights (yolo11n.pt) for transfer learning.
- Hardware Setup: GPU with 4 dedicated dataloader workers.
- Checkpointing: Best validation performance saved to weights/best.pt.

Hyperparameter	Value
Model Architecture	YOLO11n
Pretrained Weights	yolo11n.pt
Epochs	50
Batch Size	16
Hardware / Workers	GPU / 4 Workers

3.3.2 Text Detection, Recognition & Normalization





3.4 Parcel Dimension Measurement — Two Candidate Approaches

Option 1 — Webcam + Computer Vision

- Camera captures a frame; YOLO26n removes person regions and YOLO Hand Keypoints removes hand regions to prevent noise from interfering with edge detection.
- YOLO Package Detection localizes the parcel as the ROI.
- Canny edge detection scans a horizontal line at y = 0.57×h within the ROI to find left/right boundaries.
- D_cm = D_pixel × cm_per_pixel ; repeated 3× for L, W, H → V = L × W × H.

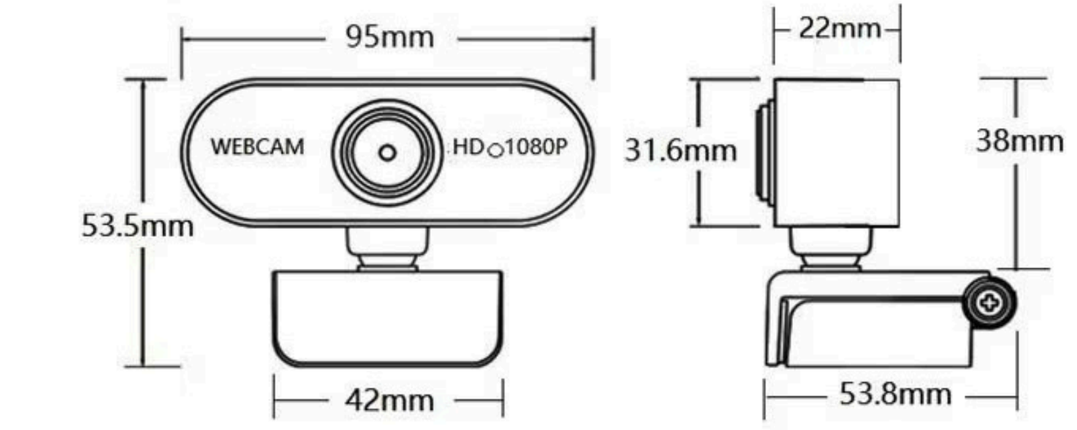
Option 2 — Ultrasonic Sensor Array + ESP32

- 3× HC-SR04 ultrasonic sensors + 1× GY-SHT30-D temperature sensor, controlled by an ESP32 microcontroller.
- Two sensors measure horizontally, one vertically (top-down), covering length, width, and height.
- Time-of-Flight principle: d_raw = v(T) × t / 2, with v(T) = 331.3 + 0.606×T.
- Each sensor is individually calibrated using linear regression to correct systematic error.

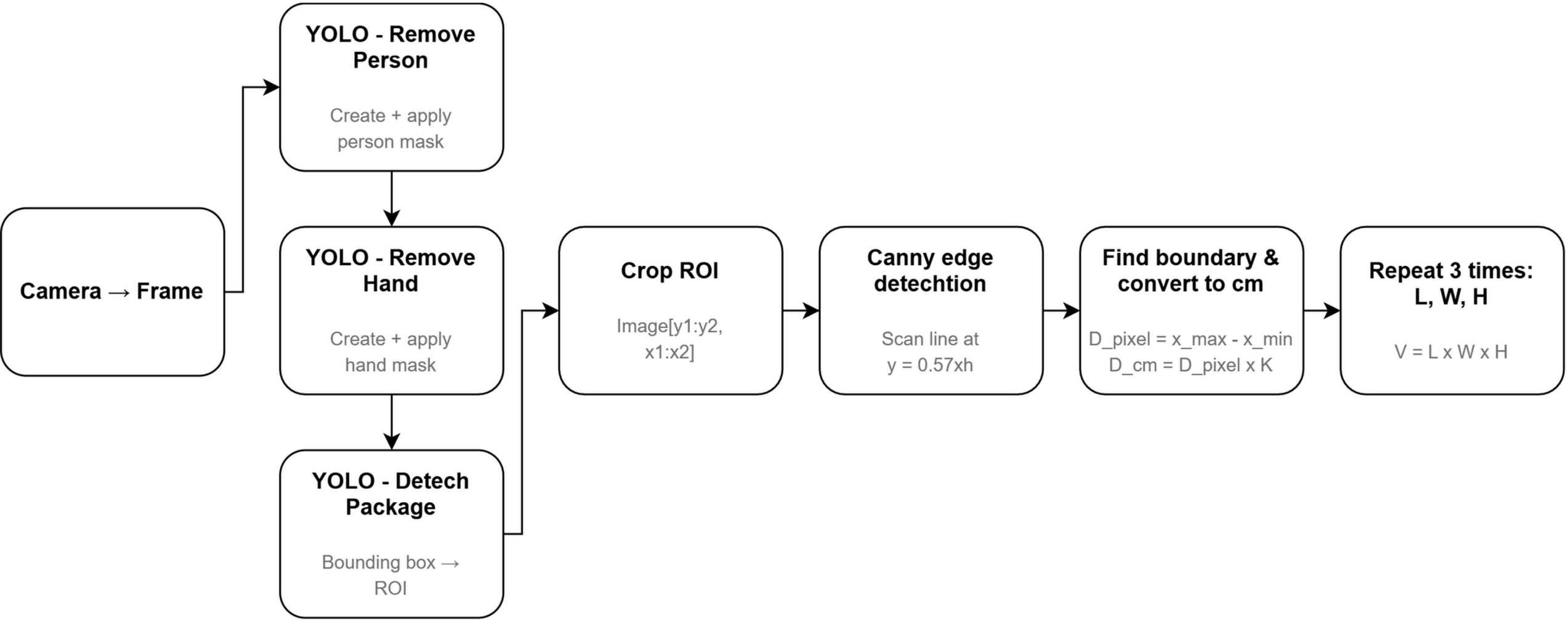
3.4.1 Webcam-Based Dimension-Measurement Pipeline

Two cameras are used to measure the volume of a package.
The first camera captures the front of the package. The second camera captures the top of the package

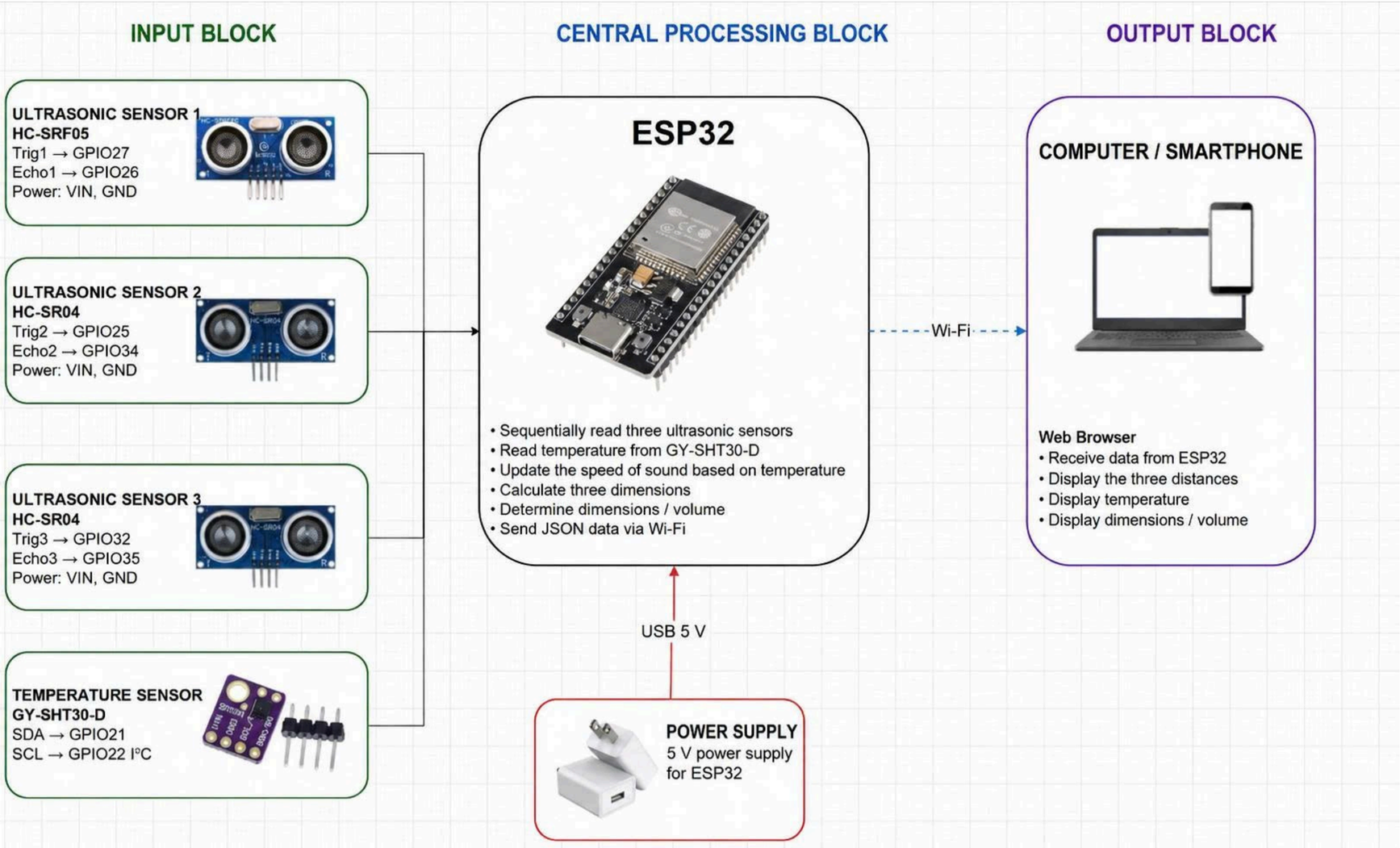




3.4.1 Webcam-Based Dimension-Measurement Pipeline



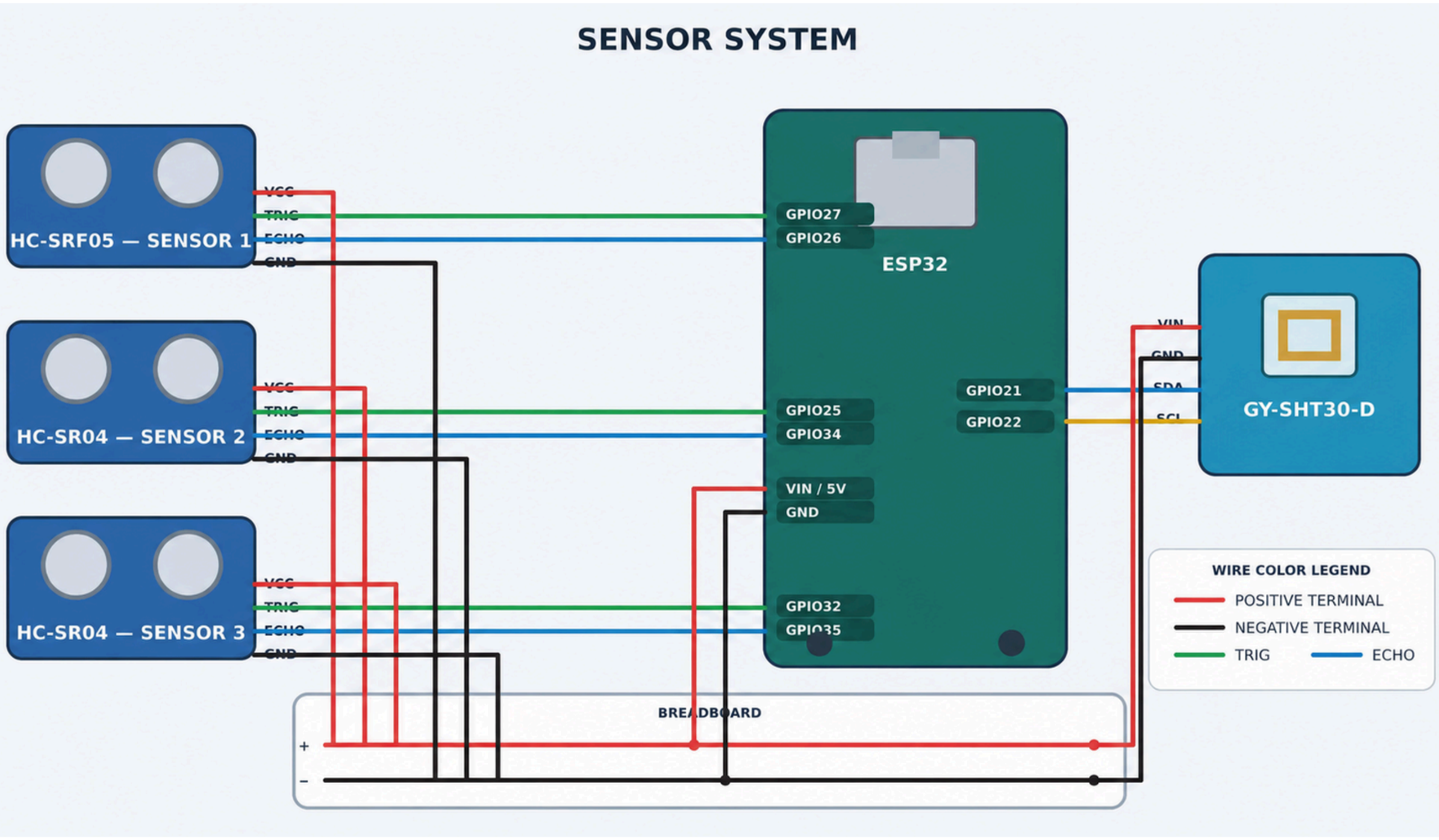
3.4.2 Ultrasonic Measurement System



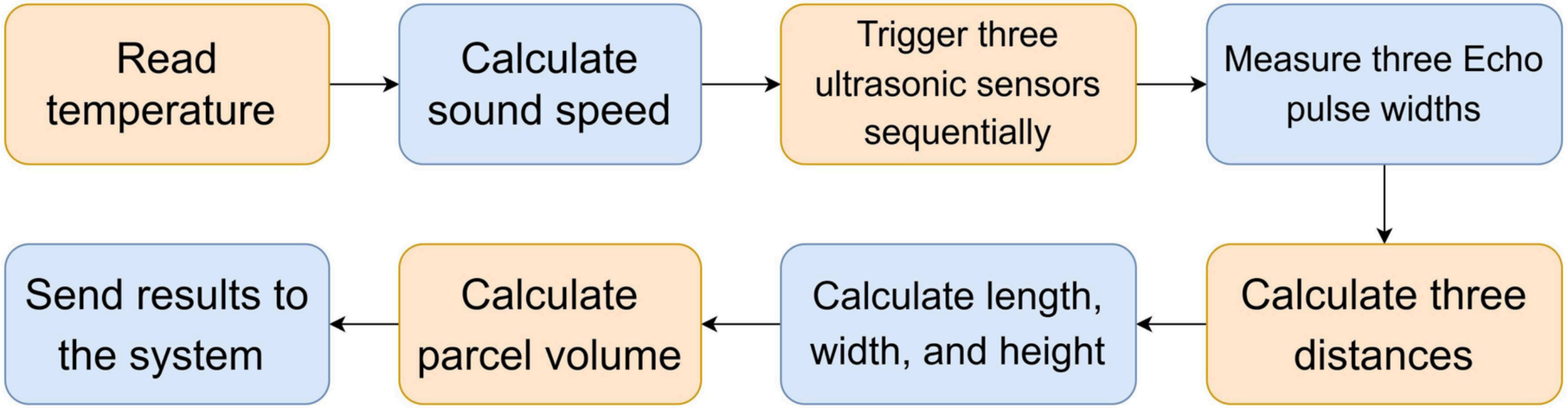
The system integrates distance and temperature measurements to calculate parcel dimensions and volume in real time.

3.4.2 Ultrasonic Measurement System





3.4.2 Ultrasonic Measurement System



Step 1:
The ESP32 reads the ambient temperature from the SHT30-D sensor and calculates the corresponding speed of sound with formula: $v(T) = 331.3 + 0.606 \times T$.

Step 2:
The ESP32 triggers the three ultrasonic sensors sequentially to reduce crosstalk. Each sensor emits a 40 kHz ultrasonic wave, while the ESP32 measures the Echo pulse width and calculates the corresponding distance.

Step 3:
The three measured distances are subtracted from the fixed reference distances to determine the parcel dimensions:
$L = 105 - d_1$ $W = 105 - d_2$ $H = 105 - d_3$

Step 4:
The parcel volume is calculated and the results are sent to the system in JSON format via Wi-Fi:
$V = L \times W \times H$

Experiments and Results

4.1 Summary of Quantitative Results — All Models

Model	Key Metrics	Result
Envelope/parcel classification (YOLOv11n)	Precision / Recall / mAP50 / mAP50-95	0.9965 / 0.9972 / 0.9950 / 0.9666
Information-region localization (YOLOv11n)	Precision / Recall / mAP50 / mAP50-95	0.9715 / 0.9524 / 0.9790 / 0.8955
Handwriting recognition (fine-tuned TrOCR)	CER / WER (test set)	5.39% / 14.99%
Person detection (YOLO26n, pre-trained)	mAP50-95	40.9
Hand detection (YOLO26n-hand, fine-tuned)	mAP50 / mAP50-95	99.5 / 89.8
Parcel detection — ROI (YOLO, fine-tuned)	mAP@50 / Precision / Recall	94.8% / 95.3% / 90.2%

4.2 Datasets

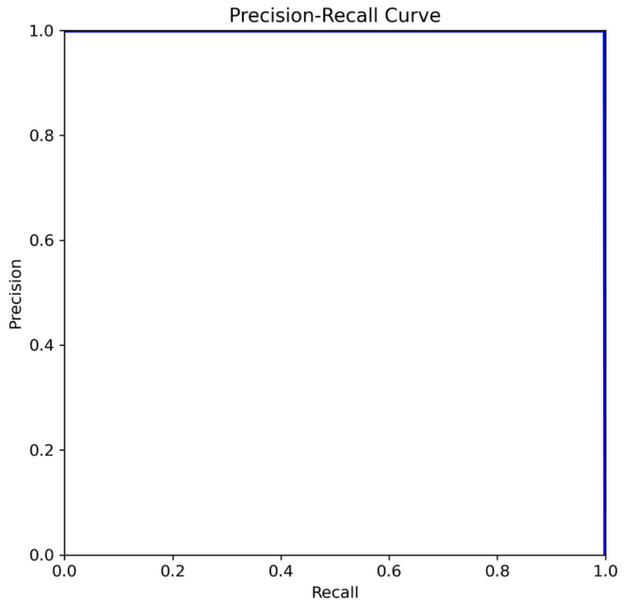
Model	Dataset information	Reference
Envelope/parcel classification (YOLOv11n)	1,137 samples (1,049 train, 44 val, 44 test)	Self-collected | Roboflow augmentation
Information-region localization (YOLOv11n)	1,137 samples (1,049 train, 44 val, 44 test)	Self-collected | Roboflow augmentation
Handwriting recognition (fine-tuned TrOCR)	Train: 59,247 samples / Test: 1,000 samples	Hugging Face
Hand detection (YOLO26n-hand, fine-tuned)	26.768 images (18.776 train/ 3.996 validation/ 3.996 test)	Ultralytics
Parcel detection — ROI (YOLO, fine-tuned)	9.216 images (7.373 train/ 922 validation/ 921 test)	RoboFlow

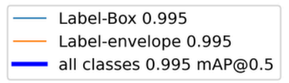
Experiments and Results

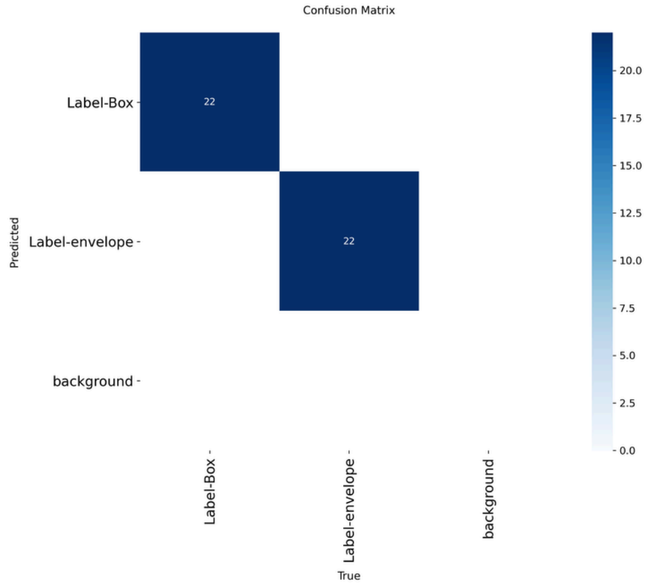
4.3 Envelope/Parcel Classification Results

0.9950	0.9666	0.9965	0.9972
mAP@0.5	mAP@0.5:0.95	Precision	Recall







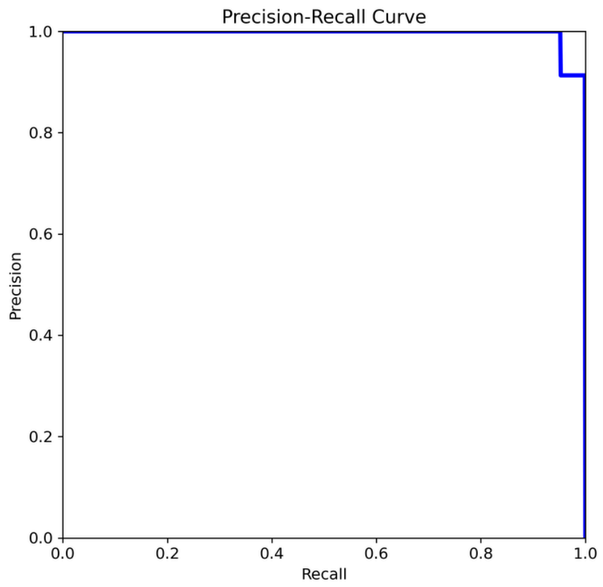


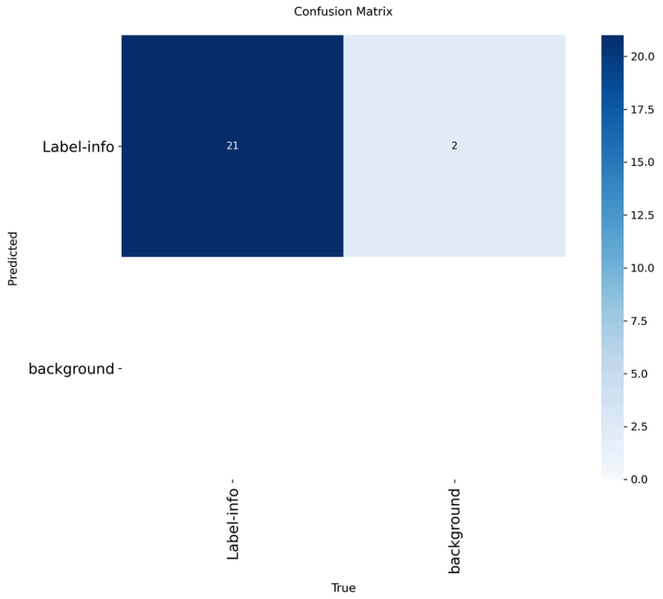
Experiments and Results

4.4 Information-region localization Results

0.9790	0.8915	0.9715	0.9524
mAP@0.5	mAP@0.5:0.95	Precision	Recall







4.5 Webcam Dimension-Measurement Results

The selection of the K-factor plays a crucial role; in many cases, while the error on individual edges may be significant, the overall volumetric error is lower than that associated with other K-factors.

Trial	Height factor	Block ratio	L (cm)	W (cm)	H (cm)	Volume error
Reference	—	—	30.00	22.00	10.00	0.00%
1	0.55·h	32/603	30.25	22.55	10.24	5.83%
4	0.55·h	32/605	30.15	22.48	10.21	4.85%
6	0.55·h	32/608	30.00	22.37	10.16	3.31%
8 (optimal)	0.55·h	32/614	29.71	22.15	10.06	0.31%

Experiments and Results

4.6 Ultrasonic Sensor Mesurement Results

Sensor	MAE before (cm)	MAE (cm)	MAPE (%)	Measurements
US1	HC-SRF05	0.9016	2.1361	9,700
US2	HC-SR04	1.7193	4.1628	9,700
US3	HC-SR04	1.7445	4.0969	9,700

- Each sensor was evaluated using 9,700 measurements collected at 97 ground-truth distances.
- US1 achieved the lowest MAE of 0.90 cm and MAPE of 2.14%.
- US2 and US3 produced MAEs of 1.72 cm and 1.74 cm, respectively.
- US1 was the most accurate sensor in the evaluated setup.
- The differences may also have been affected by sensor placement, measurement direction, and individual device variation.
- US1 achieved the lowest measurement error, which aligns with the higher specified accuracy of the HC-SRF05 sensor than the others.

05 Conclusion and Future Works

5.1 Key Finding

- The proposed multi-stage pipeline (classify → localize → detect → recognize → normalize) enables automated processing of handwritten Vietnamese postal labels.
- All four detection models (classification, localization, digit reading, parcel/person/hand detection) achieved mAP50 above 88%, confirming YOLO fine-tuning is viable for postal-specific tasks.
- Fine-tuning TrOCR on Vietnamese handwriting data sharply reduced error rates (CER 72.5%→5.39%, WER 106.95%→14.99%).
- Ultrasonic sensor calibration via linear regression cut measurement error by ~75% (CB1: MAE 0.91cm → 0.23cm).

5.2 Limitation

- End-to-end OCR accuracy (including block merging + Claude API normalization) not yet measured on large-scale real-world data.
- Digit-merging step for the digital scale module is still incomplete, blocking full end-to-end operation.
- Three-sensor ultrasonic cluster requires full hardware assembly and real-world validation.
- Webcam-based measurement depends on auxiliary models (person/hand removal) to reduce noise before edge detection.

5.3 Future works

Complete Digit Merging

- Finish the digit-to-value merging logic for the digital scale module to enable full end-to-end weight reading.

Validate Ultrasonic Hardware

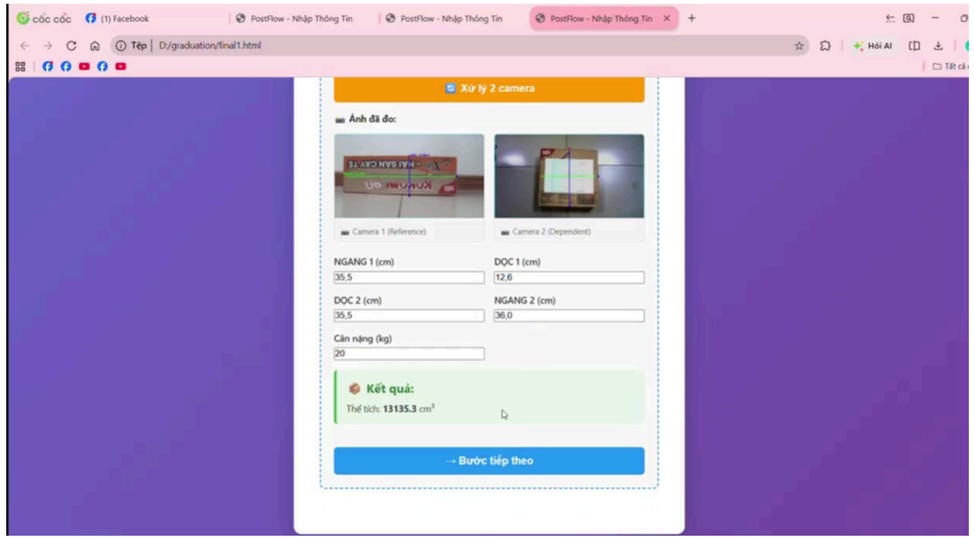
- Assemble the complete three-sensor cluster and run real-world tests to compare against the webcam-based method.

Improve Data Coverage

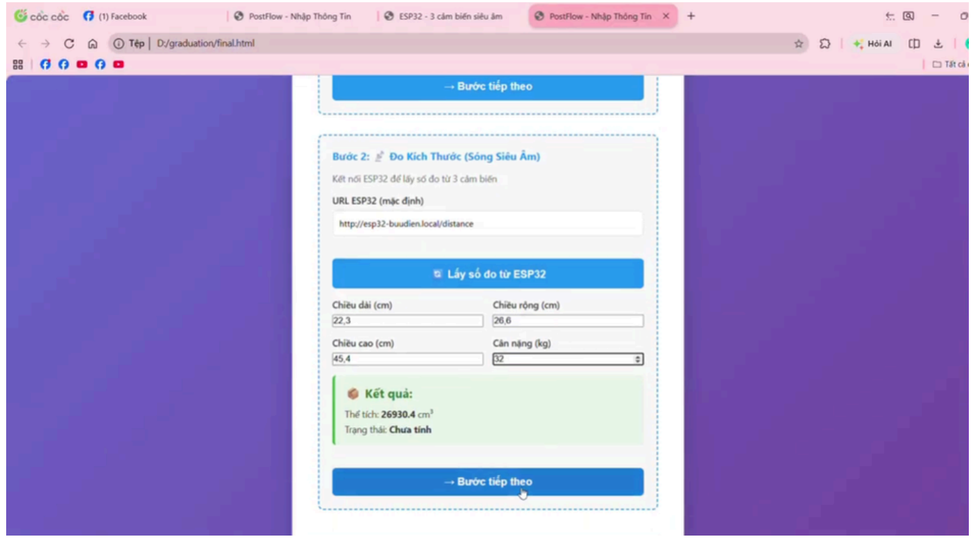
- Expand training data for information-region localization (lowest mAP50-95) and test on more diverse real-world postal labels.

Conclusion and Future Works

5.4 Demo



Automated postal system using multiple cameras



Automated postal system using multiple ultrasonic sensors

Thank You!

Questions & Discussion